FEDERAL RESERVE BANK OF MINNEAPOLIS

# QUARTERLY REVIEW

AUGUST 2025

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## Openness and Growth: A Comparison of the Experiences of China and Mexico

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Quarterly Review Vol. 45, No. 2
ISSN 0271-5287

https://doi.org/10.21034/qr.4521

This publication primarily presents economic research aimed at improving policymaking by the Federal Reserve System and other governmental authorities.

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# Tariffs, Trade Wars, and the Natural Rate of Interest*

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## Introduction

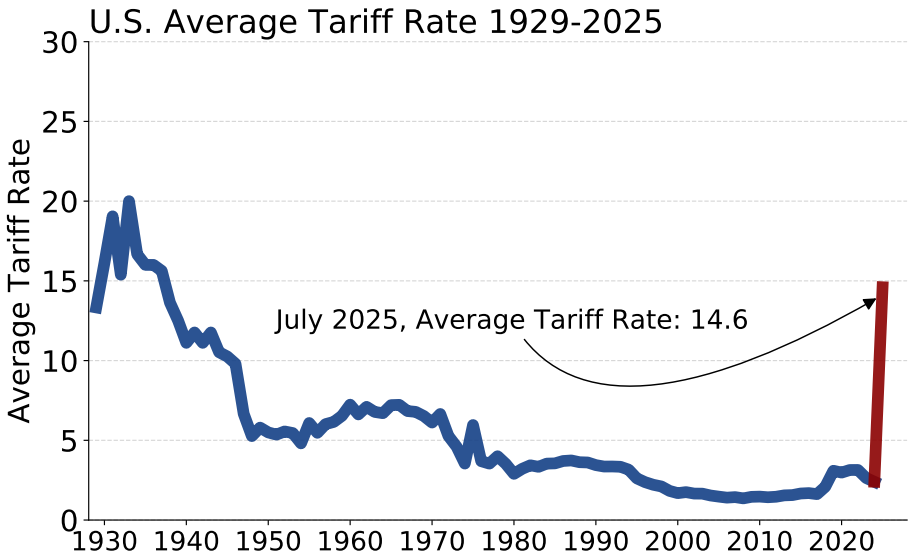
In this article, we investigate the effect of tariffs on the natural rate of interest. The natural rate is a key object for monetary policy, as it represents the theoretical neutral rate of interest, which policymakers would like to match to maintain stable inflation and full employment. Since the start of the new administration, a series of increases in tariff rates on a large number of U.S. trading partners have been implemented or announced. If these are fully implemented, the U.S. effective tariff rate would approach levels seen only a century earlier; see Figure 1. Even at current levels, the U.S. tariff rate is significantly higher than recent levels. Tariffs have effects on both aggregate demand and aggregate supply (or, alternatively, potential output). We seek to understand the latter and trace out the transition path associated with tariff increases.

Our starting point is a multicountry dynamic trade model. Households are heterogenous in the labor income earnings and their asset positions; they are subject to borrowing constraints. As a result, household asset supply is not perfectly elastic and may, in principle, be affected by tariffs even in the stationary equilibrium. A representative firm hires labor and rents capital from households. There is a single integrated global capital market, whose rental rate equates capital supply from households in all countries with capital demand from all producers.

We view this framework as a natural starting point for thinking about the natural rate of interest, because it allows for both long-run and short-run effects on the natural rate of interest via changes in both asset supply and capital demand. Moreover, much of tariff and trade policy has been justified on the grounds of persistent trade imbalances, which cannot be captured in trade models that abstract from capital accumulation or capital flows. The model we consider here represents the neoclassical benchmark.

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Figure 1
**U.S. Average Tariff Rate**




We show that the key force through which tariffs affect the interest rate is capital demand. Capital demand depends, in part, on the price of investment goods relative to the domestically produced good - the relative price of investment. To the extent that tariffs increase the relative price of investment, tariffs depress demand for capital. Because households are reluctant to lower asset supply commensurately, the decline in the demand for capital causes the natural rate of interest to fall sharply in the short run. In the long run, tariffs have only small effects on the natural rate of interest as households' asset supply falls to meet lower capital demand. Essentially, even though productivity is unchanged, tariffs appear to act like a negative productivity shock.

We calibrate the three-country (bloc) version of the model and simulate the effect of (i) a unilateral 15 percent tariff imposed by the U.S. and (ii) a global 15 percent tariff in a trade war. As with the current episode, the tariffs are announced one quarter in advance, and we examine the non-linear transition path to the new steady state.

In the unilateral tariff scenario, the tariff raises the relative price of investment and lowers the real wage in the U.S. This leads to a reduction in both capital demand and labor supply, resulting in an on-impact GDP decline of 0.6 percent. Over the long run as capital falls, a unilateral 15 percent tariff lowers GDP by nearly 1 percent.

In the unilateral tariff scenario, the natural interest rate falls by 30 basis points upon the announcement of the tariff. Both in the U.S. and the rest of the world, the relative price of investment rises, which facilitates a fall in capital demand. The natural interest rate falls to clear the capital market, and there is a resulting fall in assets in both the U.S. and in the rest of the world. The long-run effect of the tariff on interest rates is minimal, meaning that asset supply in the long run is fairly elastic and invariant to the effect of a unilateral tariff. For monetary policy, the implication is that a unilateral tariff requires a reduction in interest rates

along the transition path.

In the trade war experiment the U.S. imposes a tariff, and the E.U. and China retaliate with a 15 percent tariff against the U.S. A global trade war has large negative effects globally. The relative price of investment rises significantly, depressing capital demand. Real wages and labor supply fall globally. For the U.S., GDP declines by 1 percent on impact, and nearly 1.5 percent in the long run, as the capital stock contracts. The interest rate dynamics in a trade war are similar to the case of unilateral tariffs, but the magnitude is larger: the trade war results in a 50 basis point decline in the natural rate of interest in the short run. As in the unilateral case, the long-run effect of the tariff on interest rates is minimal and invariant to the effects from tariffs.

Our conclusion is that tariffs and trade wars are likely to lower the natural rate of interest by lowering global demand for capital. All else equal, monetary policy would need to track the lower natural rate to maintain full employment. What mechanisms could potentially reverse these conclusions?

First, our model abstracts from trade in intermediate inputs and differences between the relative price of consumption and investment goods. Higher tariffs could disproportionately raise input costs by taxing intermediate inputs or raise the replacement cost of capital. To the extent that these channels operate like anticipated productivity shocks, they would likely reinforce our conclusion that the natural rate falls and strengthen the quantitative effect. Second, our model abstracts from differences in rates of time preference. To the extent that foreign households are relatively more patient, tariffs may require higher domestic saving, which raises the natural rate of interest. Here, the assumption of a single global capital market may miss asymmetric responses in the U.S. natural rate when rates are not fully integrated. Lastly, our model may not capture differences in capital intensity across sectors or the need for higher capital investment in the U.S. to "reindustrialize" the goods-producing sector.

This paper proceeds as follows. First, we briefly describe our relation to the burgeoning literature on the macroeconomic effects of tariffs. Second, we present the model. Then, we presents our calibration and the main transition path experiments. Finally, we discuss open questions and directions for future work.

## Related Literature

The economic effect of tariffs has received considerable attention in recent months, as the Trump administration has imposed broad-based tariffs on U.S. imports. Our article is closest to Baqaee and Malmberg, which investigates the long-run effects of tariffs on the capital stock. They emphasize how tariffs raise the relative price of investment goods, and find that the "Liberation Day" tariffs, announced on April 2 (combined with commensurate retaliatory tariffs), would lower consumption by 2.6 percent; most of the effect due to lower capital accumulation. In their model, there would be no long-run effect on interest rates, but this policy would imply a large decline in interest rates along the transition path. Stepping back from tariff specifics, Ravikumar, Santacreu, and Sposi (2019) emphasize the importance of capital dynamics in trade models with changes in trade costs.

Our model builds relates to the literature on global imbalances (e.g. Caballero, Farhi, and Gourinchas 2008 and Mendoza, Quadrini, and Rios-Rull 2009) and builds on the

heterogenous agent trade framework introduced by Waugh (2023). These papers studying global imbalances have emphasized how imperfect risk sharing may lead to capital flows from financially underdeveloped to financially developed locations as households seek out safe assets. In our model, agents in each country have standard precautionary savings motives, but in the baseline calibration, we do not consider differences in the magnitude of idiosyncratic risk or introduce capital controls. The effects of tariffs on asset supply in the stationary equilibrium appear small.

Static trade models have been used to assess the costs of unilateral tariffs and trade wars but abstract from capital accumulation or interest rate dynamics (see, for example, and Simonovska et al. 2025). These models are typically richer in terms of countries and products, but the same channels are at work in our three-country framework. Optimal monetary policy with tariffs in a small open economy is considered in Bergin and Corsetti (2023), Bianchi and Coulibaly (2025), and Monacelli (2025). While we examine the dynamics of the natural rate of interest, we do not introduce sticky prices or other nominal frictions. We also abstract from more complicated rebating schemes of tariffs as emphasized in Alessandria et al. (2025).

Another line of papers has focused on how tariffs interact with trade imbalances (see, for example, Aguiar, Amador, and Fitzgerald 2025, Costinot and Werning 2025, Itskhoki and Mukhin 2025). While our model has trade imbalances, in this short article, we neither focus on them nor explore them in detail.

This article also relates to Auclert, Rognlie, and Straub (2025), which assesses the dynamic effects of a temporary shock to tariffs. These authors establish conditions under which a temporary tariff shock is recessionary and therefore lowers the natural rate of interest. Bianchi and Coulibaly (2025) establish in the calibration that the optimal monetary policy response to both temporary and permanent tariffs is to lower rates, suggesting a decline in the natural rate of interest. These models do not include capital accumulation and studies the behavior of a small open economy. By contrast, we consider both capital accumulation across countries and the global equilibrium.

## Model

This section describes the model. We first describe the actors in the economy: producers, households, the government, and global intermediaries. We then show an equilibrium of our economy, whose core element is a connection between goods flows and financial flows. We then set the stage for our quantitative results by discussing how tariffs affect capital demand and hence the equilibrium interest rate. The online appendix provides detailed derivations and characterizations of the model.

### *Production*

There are $M$ locations, each of which we call a "country." Each country produces a nationally differentiated product as in the canonical Armington model of trade. In country $i$, competitive firms have the following production technology to produce variety $i$:

$$Q_{it} = A_i K_{it}^{\alpha} N_{it}^{1-\alpha}, \tag{1}$$

where $A_i$ is total factor productivity, $K_i$ is physical capital, and $N_i$ is the efficiency units of labor supplied by households in country $i$.

Physical capital is accumulated through the acquisition of investment goods:

$$K_{it+1} = (1-\delta)K_{it} + I_{it}. \tag{2}$$

The investment good is a composite of the national varieties:

$$I_{it} = \left[\sum_{j \in M} I_{ijt}^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \tag{3}$$

where parameter $\theta$ controls the elasticity of substitution across national varieties. Associated with this aggregator is the price index $P_{I,it}$. This price index defines the number of units of the numeraire to acquire one unit of investment in country $i$. This price index also represents the value of the installed capital stock because it represents the opportunity cost of replacing capital. Thus, $P_{I,it}K_{it}$ is the value of the capital stock in country $i$.

When goods cross borders, there are two barriers to trade. The first is the standard iceberg trade costs $d_{ij} > 1$ for a good going from exporting country $j$ to importing country $i$. The second barrier is an ad valorem tariff, which is $\tau_{ijt}$. Below, we describe what the government does with the tariff revenue.

### *Households*

The household bloc builds on the standard incomplete markets model (Bewley 1979; Imrohoroğlu 1989; Huggett 1993; Aiyagari 1994) with households facing incomplete insurance against idiosyncratic productivity shocks. In each country, there is a continuum of heterogenous households $\ell \in [0,\ L_i]$; household-specific variables are denoted by superscript $\ell$. Households are immobile across countries. They are infinitely lived and have time-separable preferences over consumption and leisure:

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_{it}^{\ell} - G(n_{it}^{\ell})). \tag{4}$$

Households' period utility function is composed of several parts. One is the $u$ function, which takes the CRRA form over a consumption composite and a GHH specification for the disutility of labor supply. The consumption composite $c_{it}^{\ell}$ is a CES aggregator over the nationally differentiated varieties:

$$c_{it} = \left[\sum_{j \in M} c_{ijt}^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \tag{5}$$

where the aggregator here is the same as the one for investment goods, with $\theta$ controlling the elasticity of substitution across national varieties. When purchasing the imported varieties, the households face the same barriers to trade as with investment goods, discussed above.

Households are heterogeneous with stochastic efficiency units of labor $z_t^\ell$. These efficiency units evolve according to a standard Markov chain.

Households save in a non-state-contingent asset $a$, which is denominated in the units of the numeraire. As we discuss below, this asset is issued by competitive, global intermediaries and backed by the global capital stocks. This asset has the property that one unit of the numeraire pays out gross interest rate $R$ next period.[1]

Households can save but not borrow, and thus they face the following constraint:

$$a_{t+1}^\ell \geq 0. \tag{6}$$

These pieces come together in the household's budget constraint:

$$\sum_j p_{ijt} c_{ijt}^\ell + a_{t+1}^\ell \leq R_t a_t^\ell + w_{it} z_t^\ell n_{it}^\ell + T_{it}. \tag{7}$$

The value of asset purchases and consumption expenditures must be less than or equal to asset payments and labor earnings, and $T_{it}$ is a lump sum transfer from the government.

Let us make a couple of remarks on the role of households here. Obviously, they provide demand for the goods that other countries provide. They also play another role by saving. Individual households face idiosyncratic risk in labor earnings, and they face incomplete financial markets, thus they have a motive to save in this economy. Thus, the households are the suppliers of assets. A useful alternative approach would be to explore a setting like the one in Caballero, Farhi, and Gourinchas (2008).

### *The Government*

The government in each country merely collects the tariff revenue and rebates it to domestic households:

$$L_i T_{it} = \sum_{j \neq i} \frac{\tau_{ijt}}{(1+\tau_{ijt})} \int p_{ijt} c_{ijt}^\ell dk \; + \; \sum_{j \neq i} \frac{\tau_{ijt}}{(1+\tau_{ijt})} p_{ijt} I_{ijt}, \tag{8}$$

where the first term on the right side is tariff revenue from imported consumption goods and the second term is the tariff revenue from imported investment goods. The prices $p_{ijt}$ are the tariff-inclusive prices faced in country $i$ when importing from country $j$. The term on the left side is the total population in country $i$ multiplied by the per-capita transfer which shows up in the households' budget constraint, (7).

### *Global Intermediaries*

The final set of actors are competitive global intermediaries. These intermediaries are the owners of the capital stocks, and they receive the rents associated with the use of capital services. What this means is that the intermediary receives $r_{kt} P_{I,it} K_{it}$ from the firms in country $i$. Here, $r_{kt}$ is the global rental rate of capital of a capital service, and $P_{I,it} K_{it}$ is the value of the capital services provided in country $i$.

These intermediaries borrow from the households and pay a net return on these assets of $r_t$. To line this up with the previous paragraph, these intermediaries pay out $r_t \mathcal{A}_{it}$ to the households in country $i$ at each date $t$.

Competition implies that these intermediaries are zero profit enterprises and thus the net receipts are zero. This means that one unit of the numeraire invested in country $i$ returns $r_{kt} - \delta$ units and thus must be perfectly offset by the cost of borrowing from households at rate $r_t$. Zero profits implies that the rental rate of capital $r_{kt} - \delta = r_t$ equals the return on assets.[2]

### *Equilibrium*

This section informally discuss the conditions that an equilibrium must satisfy, with an eye toward illustrating how tariffs will affect things like interest rates and the balance of payments. The online appendix provides a formal characterization of the model and definition of an equilibrium.

First, an equilibrium is a statement about prices and quantities that are consistent with households and firms optimizing. Because this is a dynamic model, these choices must also be consistent with the laws of motion that their behavior induces.

There are two key market clearing conditions that an equilibrium must satisfy. The first is a goods market clearing condition for every country $i$ where

$$p_{iit} A_i K_{it}^{\alpha} N_{it}^{1-\alpha} = p_{iit} \int c_{iit}^k dk \; + p_{ii} I_{iit} \; + \underbrace{\sum_{j \neq i} \frac{p_{jit}}{1+\tau_{jit}} \int c_{jit}^k dk + \sum_{j \neq i} \frac{p_{jit}}{1+\tau_{jit}} I_{jit}}_{\text{Exports}_{it}} . \tag{9}$$

This says that value of production must equal the value of domestic consumption and investment goods plus the exports of consumption and investment goods, excluding the foreign tariffs.

This equilibrium condition can also be reframed into a balance of payments (BOP) accounting identity. The BOP identity is useful because it illustrates how trade balances evolve in this economy. Starting with (9), the online appendix shows how one can derive the BOP identity:

$$\underbrace{\underbrace{\Big[\text{Exports}_{it} \; - \; \text{Imports}_{it}\Big]}_{\text{Trade Deficit}} + \underbrace{\Big[r_t(\mathcal{A}_{it} - P_{I,it} K_{it})\Big]}_{\text{NFI}}}_{\text{Current Account}} = \underbrace{\Big[(\mathcal{A}_{it+1} - \mathcal{A}_{it}) - P_{I,it}(K_{it+1} - K_{it})\Big]}_{\Delta\text{NFA}}, \tag{10}$$

where NFI is net foreign income and $\Delta$NFA is the change in net foreign asset position. Let us make a couple of observations. In a stationary equilibrium, a country's asset supply $\mathcal{A}_{it}$ and capital demand $P_{I,it} K_{it}$ do not change; thus, thus the change in net foreign assets is zero, and current accounts are zero. This *does not* imply that trade deficits are zero, as a country's

asset supply $\mathcal{A}_{it}$ may not match up with capital demand, and hence net foreign income (and trade deficits) may be positive or negative. Along the transition path, the net foreign assets position will be changing as a country's asset supply and capital demand adjust—say in response to a tariff. While simple, this model does nicely integrate ideas about trade in both goods and assets that allow us to take a first pass at thinking about tariffs and interest rates.

The second important equilibrium condition is the global asset market clearing condition. This condition requires that

$$\sum_{i \in M} \mathcal{A}_{it+1} = \sum_{i \in M} P_{I,it+1} K_{it+1}; \tag{11}$$

that is, the global value of asset holdings must equal the global value of the capital stocks.

The principle price moving to satisfy (11) is the global interest rate. This interest rate dictates savings behavior in all countries and thus the left hand side of (11). This interest rate also dictates capital demand and thus the right side of (11). What we term the natural rate of interest is the rate that matches asset supply and capital demand in the global economy. Below, we focus on how tariffs would affect capital demand and, in turn, the prevailing interest rate.

### *A Discussion of Tariffs, Capital Demand, and Interest Rates*

How will tariffs can affect interest rates? The key issue is that a tariff increases the relative price of investment and thereby lowers the demand for capital. The interest rate must then adjust to induce a reduction in the supply of assets by households and / or offset the relative price effect for the demand for capital.

To understand this logic, consider the demand for capital:

$$K_{it+1} = \left[\alpha(r_{t+1} - \delta)\right]^{\frac{1}{\alpha-1}} \left[\frac{P_{I,it+1}}{p_{ii,t+1}}\right]^{\frac{1}{\alpha-1}} N_{it+1}, \tag{12}$$

which derives from the first-order condition for a firm in country $i$. Several prices modulate the demand for capital. A higher interest rate reduces capital demand because this rate determines the cost of renting out capital services. A higher relative price of investment reduces capital demand as well. The relative price reflects the cost of using capital services (through $P_{I,it+1}$) versus the value of adding an additional unit of capital (through the $p_{iit}$ term).

Our argument is that the direct effect of a tariff is to increase the relative price of investment (as long as a tariff reduces trade). To see this more clearly, we can use the logic of Arkolakis, Costinot, and Rodríguez-Clare (2012) to connect relative prices with trade flows through the CES demand system:

$$K_{it+1} = \underbrace{\left[\alpha(r_{t+1} - \delta)\right]^{\frac{1}{\alpha-1}}}_{\text{interest rates}} \times \underbrace{\left[\pi_{iit+1}\right]^{\frac{1}{(\theta-1)(\alpha-1)}}}_{\text{trade}} \times \underbrace{N_{it+1}}_{\text{labor supply}}, \tag{13}$$

where the center term $\pi_{iit+1}$ is the "home trade share," the share of total expenditures spent on goods from home. A tariff shifts expenditures toward domestically produced goods and away from foreign produced goods; hence, the home share $\pi_{iit+1}$ increases. With the standard parameter restrictions that $\alpha < 1$ and $\theta$ is positive, a tariff decreases capital demand. And the strength of these effects is modulated by the importance of capital in production and the trade elasticity.

It's worth looking at (11) and (13) to think through how an equilibrium must adjust when a tariff reduces capital demand. One important force is that the interest rate will decrease. A lower interest rate does two things to restore an equilibrium. From the households' problem, a decrease in interest rates will support a reduction in asset supply as households find the intertemporal trade-off of consumption today versus tomorrow less favorable. Second, a lower interest rate also offsets the higher relative price of investment and is a force to increase capital demand. In the numerical exercise below, we find that a tariff lowers the natural rate of interest in the short run.

To be clear, there are other forces operating in the background as well. An increase in the price index of consumption lowers the real wage, which lowers labor supply, and through (13), would act as a force to reduce capital demand. Over the long run, a country has less capital, real wages are lower, and households are ultimately poorer. This implies that asset supply will fall as well.

Finally, the asset market clearing condition in (11) is global, and an obvious question is the extent to which other countries in the world can make up for the reduction in the demand for capital in the tariff imposing country. Moreover, the logic put forth above is stronger when one thinks about a global trade war. For all countries, their relative price of investment will fall, which will reduce global demand for capital. This will require a fall in interest rates to support a reduction in asset supply.

Conceptually, these possible effects are straightforward to understand. However, what is not clear is their timing and quantitative magnitudes, especially with offsetting effects from other countries. Thus, we turn to a numerical example next to illustrate the possibilities.

## How Do Tariffs Affect Interest Rates?

In this section we describe the quantitative results from the impact of a unilateral tariff and a trade war. First, we briefly discuss the calibration approach. We then present the results of studying the nonlinear transition paths to changes in trade policy.

### *Calibration*

Our intent here is to be illustrative, and hence we take a simple approach to calibrating the model. In the final section, we discuss open questions, some of which concern the inputs into our numerical exercise.

The time period in the economy is a quarter. Our economy will consist of three symmetric countries. Think of our three countries as being the U.S., the E.U. bloc, and China. These three countries / blocs are identical in terms of population, preferences, the stochastic process for labor productivity, aggregate total factor productivity, and iceberg trade costs.

Preferences in all countries are of the $\log$ form for the utility function and the GHH iso-elastic form for the disutility of labor $G(n)$. We set the scale of the disutility of labor equal to a household's labor productivity; consequently, all households within a country supply labor as a function of only the real wage per efficiency unit. We set the labor supply elasticity parameter to $0.50$, which is a standard value.

We set the value of the trade elasticity to 4 which is consistent with the evidence in Simonovska and Waugh (2014).

We set the depreciation rate to be 10 percent and then choose the capital share parameter $\alpha$ to return a capital to output ratio of 2.5; the resulting capital share is 0.26. Simultaneously, the discount factor is chosen to target an annual interest rate of 1.5 percent, which is consistent with recent estimates of the natural rate of interest over past 7 years (Federal Reserve Bank of New York 2025).

Finally, the iceberg trade costs are set to target a home trade share $\pi_{iit}$ for all countries at about 85 percent. This is consistent with an imports / GDP ratio in 2024 for the U.S. and China of around 14 percent and 18 percent respectively. It does understate the openness of the E.U. bloc. Tariffs in the initial stationary equilibrium are zero.

### *The Tariff Shock*

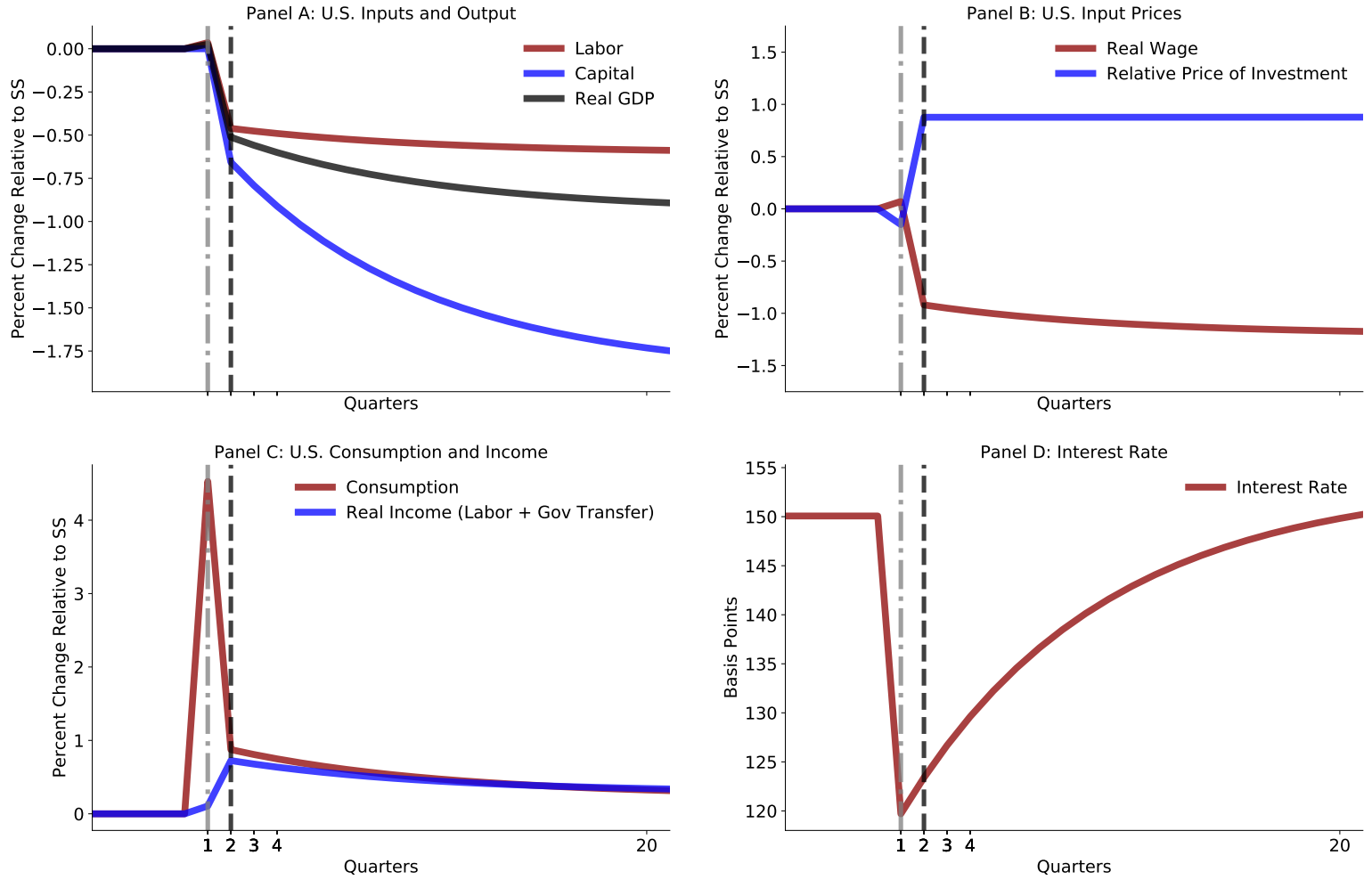
Given the calibrated model, we then shock the economy with a surprise imposition of tariffs and study how the economy responds along the transition path to the new stationary equilibrium.

As mentioned above, two tariff exercises are considered. The first one is where the U.S. imposes a unilateral tariff on its trading partners. The size of the tariff that we consider is 15 percent. This number is consistent with measures of the effective tariff rate imposed by the Trump administration through July 2025 (see, e.g., www.tradewartracker.com 2025). We also provide actors in the economy with some news about this shock. Similarly to how the tariffs were rolled out in the first half of 2025, one quarter before the tariff is imposed, all actors in the economy receive advance notice. Figure 2 presents the results for the unilateral case

The second exercise is a trade war where the U.S. imposes a tariff on its partners (the E.U. and China) and the other countries impose a tariff of equivalent size on the U.S. There are no tariff actions between the E.U. and China. As was the case above, the size of the tariff is 15 percent, and tariffs are announced one quarter in advance. Figure 3 presents these results.

Figure 2

**U.S. unilateral tariff**



### *U.S. Unilateral Tariff*

In Figure 2, there are four panels. They describe (i) how inputs and output respond in the U.S., (ii) how measures of relative prices respond, (iii) the response of consumption and real income (which is labor earnings plus government transfers), and (iv) the global interest rate. The vertical dash-dot line, at quarter 1, is when the tariff is announced, the vertical dashed line, at quarter 2 is when the tariff is put in place.

Panel A of Figure 2 shows how a tariff reduces the use of capital and labor and this leads to a loss in real GDP for the U.S.[3] The output losses are meaningful: real GDP falls by about half of a percent in the quarter after the tariff is in place and one percent in the long-run.

The reason for the decline in capital follows from the section above about the relationship between tariffs and capital demand. The tariff increases the relative price of investment, and this reduces capital demand in the U.S. Panel B illustrates this point. It shows how the U.S. relative price of investment increases by about 1 percent when the tariff goes into place and then stays essentially flat throughout the period. The other channel through which capital demand is depressed is labor supply. The tariff results in a fall in the U.S. real wage by about 1 percent when the tariff goes into place.[4] U.S. households respond by working less, and in turn, American firms demand even less capital.

The fall in capital demand is met by a rise in U.S. consumption. Panel C displays the dynamics of consumption and real non-asset income. In the period just before the tariff is put in place, consumption increases substantially, even though real income is essentially unchanged. This means that U.S. households are aggressively consuming out of their asset income and saving less.

There are several forces behind this behavior. One is that the price index of consumption increases in quarter 2, and thus households pull forward consumption into quarter 1. before the implementation of the tariff. Second, there is also a sharp fall in the global interest rate (Panel D), which induces U.S. households to consume more. In the quarter when the tariff is announced, there is a 30 basis point drop in the interest rate, and it slowly rises over the next 5 years, returning to a point slightly higher than the initial steady state.

Globally, similar forces are at play, which reinforce the effects on the interest rate. In the E.U. and China, the relative price of investment rises although by a smaller amount than in the U.S. The real wage falls as well, reducing labor supply. On net, these forces are driving a reduction in demand for capital, and importantly, other countries cannot make up for lost capital demand in the U.S. For households in these countries, consumption falls, but this fall is substantially less than that of households' real income. Thus, the fall in the global interest rate encourages a global reduction in asset supply to line up with global reduction in capital demand.

### *Trade War*

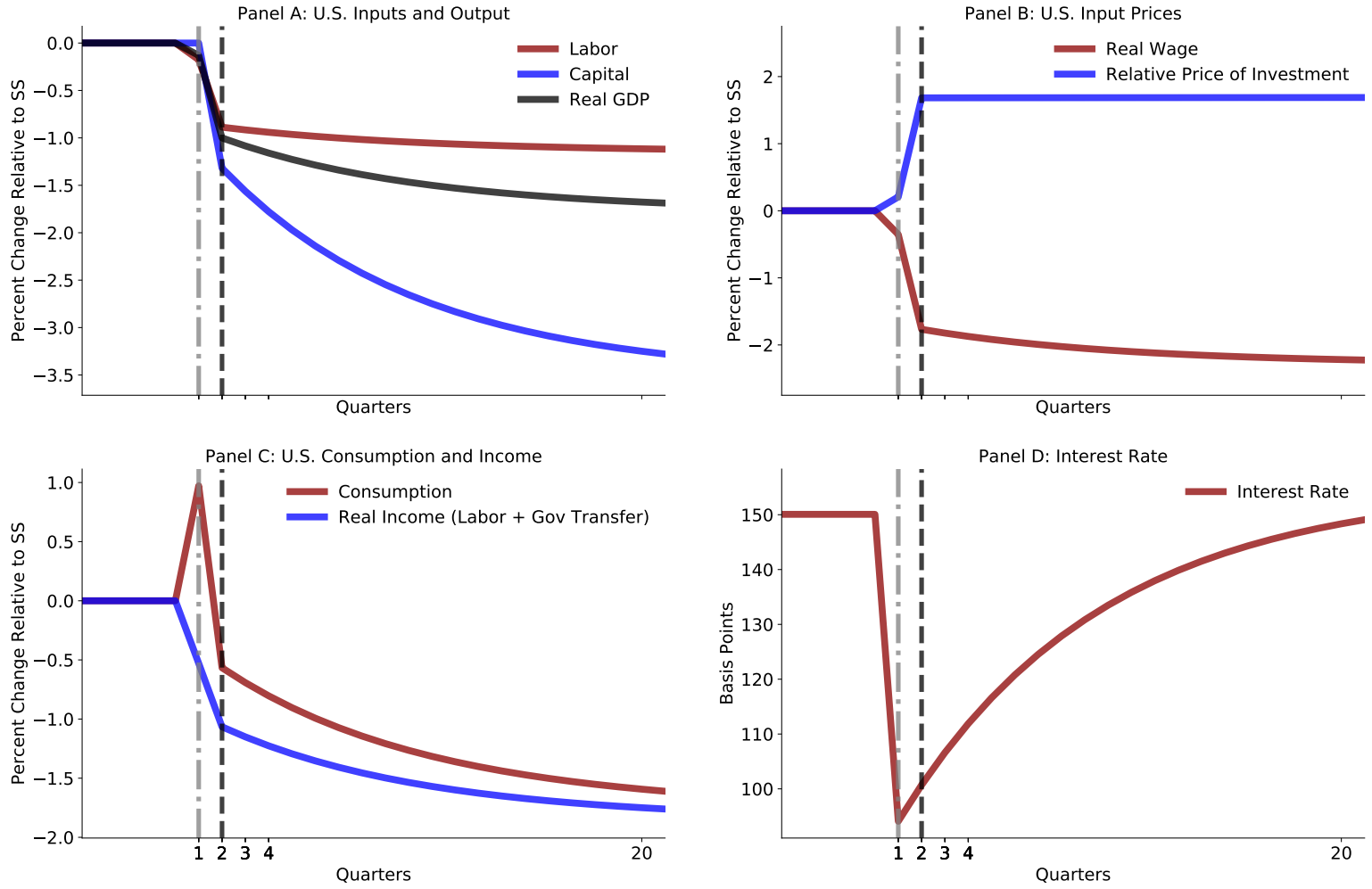
Figure 3 presents the results for the trade war. The format of Figure 3 is the same as the one in in the previous figure.

As with a unilateral tariff, in a trade war, the demand for both capital and labor falls in the U.S. However, the quantitative magnitudes are much larger. Capital and labor decline by a factor of two, as does real GDP in both the short and the long run. And the relative price of

investment and real wages follow the same pattern as the unilateral tariff scenario. As Panel B of Figure 3 shows, the relative price of investment increases, and the real wage falls, in both cases by substantial amounts. The result is that U.S. households respond by working even less, relative to the unilateral tariff case. With a higher price of investment and fewer workers, American firms demand even less capital.

As with the unilateral tariff scenario, there is a temporary boom in real consumption before the tariff is implemented (Panel C of Figure 3). The important difference is that consumption falls rapidly after the tariff is implemented and by large amounts, half a percent on impact and nearly 1.5 percent in the long run. What is important to notice is that consumption is falling by *less* than the decline in real income. This means that U.S. households are consuming out of their asset income and saving less. The economics are similar to those in the unilateral tariff scenario. U.S. households see the consumption price index rising in quarter 2 and pull forward consumption. But their future balance sheets are overall weaker, and this force is not as strong as in the unilateral case.

Figure 3
**Trade war**



The second reason consumption falls in a trade war is the larger decline in the global interest rate. Panel D of Figure 3 illustrates this point: the announcement of the trade war triggers a 50 basis point drop in the interest rate. The fall in the interest rate induces U.S. households to run down their savings and supply less assets.

Similar effects take place in the other countries. However, the magnitudes for other countries are not as large as those for the U.S. as the other countries retain the ability to trade with each other tariff-free. In all countries, the relative price of investment rises, the real wage declines, and capital demand falls. For households in these countries, consumption falls, but by less than the fall in real income, and thus foreign households are saving less and supplying fewer assets. And the tariff-induced fall in the global interest rate is what induces a reduction in asset supply to meet the tariff-induced reduction in capital demand.

## Discussion

As we emphasized in the introduction, the model we consider here represents a starting point for thinking about the effects of tariffs on the interest rate and return on capital. There are multiple dimensions along which we could extend the model to better capture the channels through which tariffs impact current and future productivity. Our model does not capture trade in intermediate goods, raw materials, or capital goods. More than two-thirds of U.S. trade consists of trade in either intermediate or capital goods. Tariffs, by raising the relative price of inputs to production, may have stronger effects on current and future productivity than what our model currently captures. We conjecture that explicitly incorporating these channels would likely magnify the impact on future productivity and the decline in the natural rate of interest in the short-run.

Imperfect capital mobility and global imbalances are more likely to potentially attenuate or reverse the effect of tariffs on the natural rate. Much of the policy discussion has emphasized the size of the U.S. trade deficit and the deindustrialization of the U.S. manufacturing base. Persistent trade deficits may reflect that U.S. households and businesses are less patient than their foreign counterparts and are borrowing from foreigners to finance high levels of current consumption, relative to income. It seems possible that tariffs | in order to lower domestic consumption relative to income | may induce a higher natural rate of interest in the domestic economy. Here, we are reasoning from two-agent models along the lines of Mehrotra (2018) or Eggertsson and Krugman (2012), where increased financial frictions or deleveraging results in implicitly higher rates for borrowers.

We suspect that a reindustrialization channel may likewise raise the natural rate of interest after the imposition of tariffs. Globalization has led to a migration of capital intensive production from the U.S. and other advanced economies toward East Asia and China, in particular. The imposition of tariffs may require a reallocation of production from services toward goods, which would raise the marginal product of capital in goods production. This reallocation effect may attenuate or reverse the lower productivity effect of tariffs and raise the natural interest rate in the U.S. To our knowledge, neither the qualitative nor the quantitative importance of this effect has been investigated. It represents a clear direction for future work. Importantly, intangible capital investments have been important for information and financial services - industries in which the US is a net exporter of services. Tariffs may *negatively* drive down the marginal product in those industries, offsetting any

increases in goods production.

Overall, we think further work is needed to create quantitative models that capture the key static and dynamic elements of trade: 1) trade in intermediates and capital goods, 2) persistent global imbalances in capital flows, and 3) sectoral reallocations between manufacturing and services. Tariffs and trade policy changes will likely have sizable effects across all these channels, and both the direction and magnitude of these effects are important directions for future work.

## Notes

1. What is the numeraire? We keep the numeraire in abstract units that are not intrinsically valued by households. And we do so by setting it equal one unit of global consumption (which means the average of the consumption price indexes are set to one). Think of this abstract unit like a SDR.

2. This structure is not critical. For example, having firms issue bonds and / or households have direct ownership of the capital stocks would lead to the condition $r_{kt} - \delta = r_t$. What this structure does do is it place ultimate ownership of the capital stocks in these intermediaries' hands, and thus the model is informative only about net asset positions, not gross asset positions.

3. We measure real GDP as physical output valued at the initial steady state prices. This is *not* the value of output deflated by the consumption price index. Thus, any move in real GDP is of changes in factors (labor and capital), not changes in the relative value of output, which is what the later measure would reflect.

4. Here "real" means wages deflated by the contemporaneous consumption price index. This measure is closer to the welfare metric used in static trade models like the one in Arkolakis, Costinot, and Rodríguez-Clare (2012).

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